IBAC 2025
KERTEMINDE | DENMARK

SDU
University of Southern Denmark

S1-1

Do nonlinear phenomena in dog puppy whines signal distress?

Mathilde Massenet, Andrey Anikin, Katarzyna Pisanski, Romane Philippe, Karine Reynaud, Nicolas Mathevon, David Reby

Affiliation of presenting author:

Department of Communica8on, University of California Los Angeles, USA ENES Bioacous8cs Laboratory, University of Saint-E8enne, France

Domestic dog puppies typically produce whines in contexts associated with distress, such as separation from their mothers. Like the cries of human and nonhuman infants, whines are highpitched and characterized by a harsh-sounding vocal quality due to the presence of nonlinear phenomena (NLP). While whines are known to attract the attention of dog mothers and human caregivers, the extent to which NLP convey distress information remains unclear. We conducted a series of experiments investigating the production and perception of NLP in puppy whines. Specifically, we used acoustic recordings of whines produced by 3-week-old Beagle puppies experiencing brief separation from their mother for playback experiments involving dog mothers and human listeners. The production of whines containing NLP increased with time since separation, and dog mothers inspected their puppies for longer when exposed to synthetic whines with experimentally added NLP compared to whines without NLP. Additionally, human listeners rated synthetic puppy whines with NLP as indicating greater distress compared to those without NLP. Our results provide strong evidence that NLP in puppy whines are critical for signaling distress and therefore may be crucial for offspring survival during breeding in a domesticated species. These findings may also contribute to the design of bioacoustics-based tools for monitoring animal welfare.

S1-2

The hidden complexity of African Penguin begging calls: Nonlinear Vocal Phenomena and their significance

Livio Favaro, Zanoli A., Morandi I., Terranova, F., Tenneriello C., Balestra F., Monaci L., Marmoreo V., Cadman M., Ludynia K.

Affiliation of presenting author:

Department of Life Sciences and Systems Biology, University of Turin, Turin, Italy

African penguin (Spheniscus demersus) chicks use high-pitched begging calls to stimulate food regurgitation from their parents. We investigated the occurrence of Nonlinear Vocal Phenomena (NLP) in these food solicitation signals of both hand-reared birds at the Southern African Foundation for the Conservation of Coastal Birds and wild chicks at the Stony Point Penguin Colony in South Africa. For each chick, we recorded begging calls from the early days after hatching for approximately three months while also monitoring their age as an indicator of growth and assessing their health condition. The visual inspection of spectrograms by independent observers revealed that most (>60%) begging calls of healthy chicks contained NLP, with sidebands and deterministic chaos being the most frequently observed patterns, often co-occurring within the same call (>26%). We propose that these aperiodic chaotic features of food solicitation signals may help increase adult attention and prevent habituation. However, our findings also indicate a significant increase (up to 80%) in NLP occurrence in chicks contracting severe respiratory diseases (i.e., bacterial infections or aspergillosis). Importantly, when these chicks received antimicrobial treatment, their begging calls exhibited a significant reduction in NLP, reaching values comparable to those of healthy individuals. Building on this, we trained Convolutional Neural Network models on annotated datasets to detect NLP in non-annotated sequences of begging calls. Automatically estimating the NLP occurrence threshold might assist in identifying and diagnosing individuals requiring veterinary care, thereby contributing to conservation efforts for this Critically Endangered species.

S1-3

‘Monkey yodels’—frequency jumps in New World monkey vocalizations greatly surpass human vocal register transitions

Jacob C. Dunn, Christian T. Herbst, Isao T. Tokuda, Takeshi Nishimura, Sten Ternström, Vicky Ossio, Marcelo Levy, W. Tecumseh Fitch

Affiliation of presenting author:

Behavioural Ecology Research Group, Anglia Ruskin University, UK; ENES Bioacoustics Research Laboratory, University of Saint-Etienne, France; Department of Behavioral and Cognitive Biology, University of Vienna, Austria

Vocal membranes – thin upward projections of the vocal folds – are typical features of the primate larynx, but were lost in human evolution as a presumed adaptation for speech. In primates, the vocal membranes are active during phonation, along with the vocal folds. While their precise acoustic function remains poorly studied, vocal membranes are thought to play a crucial role in generating the high proportions of nonlinear phenomena observed in primate vocalisation. Here, we investigated the causal basis of abrupt frequency jumps in a unique database of New World monkey vocalizations. We used a combination of acoustic and electroglottographic recordings in vivo, excised larynx investigation of vocal fold dynamics, and computational modelling. We observed two distinct modes of phonation. The first, involving vocal fold vibration alone, produced low-frequency oscillations, and is analogous to that underlying human phonation. The second, incorporating the vocal membranes, resulted in much higher-frequency oscillation. Abrupt frequency jumps between high and low frequencies were systematically present in our data, and were causally driven by switching between these distinct laryngeal mechanisms. While these data are reminiscent of the rapid transitions in frequency observed in certain human singing styles (e.g., yodelling), the frequency jumps are considerably larger in primates. Our data suggest that peripheral modifications of vocal anatomy provide an important source of variability and complexity in the vocal repertoires of nonhuman primates. We further propose that the call repertoire is crucially related to a species’ ability to vocalize with different laryngeal mechanisms, analogous to human vocal registers.

S1-4

Non-Linear phenomena in bonobo vocalizations: A key acoustic feature for emotional communication across species

Fournier Floriane, Leo Perrier, Sumir Keenan, Amanda Epping, Jared Taglialatela, Celle Laurent, Katarzyna Pisanski, Florence Levréro

Affiliation of presenting author:

ENES Bioacoustics Research Laboratory, ENES/CRNL, University of Saint-Etienne, CNRS, Inserm, Saint-Etienne, France

The perception and interpretation of emotional expressions play a crucial role in regulating both intra- and interspecific interactions, potentially leading to emotional contagion or more complex forms of empathy between and across species. Previous research has shown that the level of emotional arousal and valence influences the acoustic features of vocalizations. These acoustic variations are not only perceived within species but also across species. Notably, nonlinear phenomena (NLPs)—such as subharmonics, sidebands, deterministic chaos, and frequency jumps—have been linked to emotional intensity in various mammals, including primates. To investigate the potential effect of NLP, we recorded bonobo vocalizations across different contexts and quantified the presence of NLPs. We then resynthesized these vocalizations by incorporating sidebands before playing them back to both bonobos and humans. We monitor the physiological responses of both humans and bonobos using ECG devices as well as the behavioural response of bonobos. We found in bonobos (Pan paniscus) that NLPs are more prevalent in vocalizations associated with high arousal and negative valence than low arousal and positive valence. Furthermore, our results show that vocalizations artificially enriched with sidebands can trigger physiological changes in both bonobos and humans, as well as behavioural changes in bonobos. This cross-species finding highlights the evolutionary significance of NLPs, suggesting that their presence may lead to heightened arousal in perception and play a fundamental role in emotional communication.

S1-5

The cultural evolution of distortion in music (and other norms of mixed appeal)

Greg Bryant, Paul Smaldino

Affiliation of presenting author:

Department of Communication, University of California, Los Angeles, CA, USA.

Music traditions worldwide are subject to remarkable diversity but the origins of this variation are not well understood. Musical behavior is the product of a multicomponent collection of abilities, some possibly evolved for music but most derived from traits serving nonmusical functions. Cultural evolution has stitched together these systems, generating variable normative practices across cultures and musical genres. Here I will describe the cultural evolution of musical distortion, a noisy manipulation of instrumental and vocal timbre that emulates nonlinear phenomena (NLP) present in the vocal signals of many animals. Listeners' sensitivity to NLP has facilitated technological developments for altering musical instruments and singing with distortion, which continues to evolve culturally via the need for groups to both coordinate internally and differentiate themselves from other groups. To support this idea, I'll present an agent-based model of norm evolution illustrating possible dynamics of continuous traits such as timbral distortion in music, dependent on (i) a functional optimum, (ii) intra-group cohesion and inter-group differentiation and (iii) groupishness for assortment and social learning. This account illustrates how cultural transmission dynamics can lead to diversity in musical sounds and genres, and also provides a more general explanation for the emergence of subgroup-differentiating norms.

S1-6

Biphonation in animal vocalizations: from communicative functions to production mechanisms

Romain A. Lefèvre, Océane Amichaud, Doğa Özcan, Elodie F. Briefer

Affiliation of presenting author:

Behavioural Ecology Group, Section for Ecology & Evolution, Department of Biology, University of Copenhagen, 2100 Copenhagen Ø, Denmark

Biphonation is a 'nonlinear phenomena' defined as the simultaneous production of two distinct, non-harmonically related fundamental frequencies. It has traditionally been viewed as an anomaly or a by-product of vocal pathology. However, recent studies have challenged this assumption, showing that biphonic calls are present in the natural vocalizations of a wide range of taxa, including birds, amphibians and mammals. This talk will review the potential communicative functions of biphonic calls, and their possible production mechanisms. Biphonation could play an essential role in communicating distinct pieces of information at short- versus long-distance, increase call complexity to allow more individually distinct calls, and provide cues to the sender's direction of movement. Proposed mechanisms underlying biphonation production include asymmetries in vocal fold oscillations, the addition of aerodynamic whistles, the involvement of secondary structures, and bilateral specializations. Overall, the current evidence suggests that biphonation constitute a significant phenomenon within the acoustic repertoires of various species.